Lab

Java basics — Arrays and loops

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Lab: Loops and arrays

Objective

The objectives of this practical are:

- To get fully up to speed with array definition, creating, filling, and manipulation.
- To fully understand when it is appropriate to use a ‘foreach’ loop as opposed to a ‘for’ loop to process the array.

Part 1 — Getting started!

1. Back in the **labs** project, created a new class called ***Program*** in a package called lab **lab4** with a main() method.
2. Declare an array of integers in the main() method as:
   int[ ] numbers = { 1, 3, -5, 7, 0, 4, 6, 8 };
   Then, without using built-in commands in Java, do these tasks:

- **Task 1:** Write code to find the sum of every number in numbers
- **Task 2:** Find the average of these numbers
- **Task 3:** Find the minimum number in numbers
- **Task 4:** Find the maximum number in numbers
- **Task 5:** Find the index of number zero in numbers

Part 2 — Calculating the grades for five students

Step-by-step

1. We will revisit the grades() method that you wrote before but this time we will process many grades rather than just one.
2. Create a method called **getGrade** like:

   String getGrade(**int** mark)
3. Copy the code for processing grades to this method.
4. Back in the main() method, create an array of five names (String[ ]) called **students** and then create an array of five integers called **marks** to hold the marks for our five students.

   Fill the students[ ] array with five names and the marks[ ] array with five marks.
5. Use a **for** or **while** loop to determine the grade for each mark in the marks array and display it alongside the corresponding student name from the students array.

Part 3 — How long does it take to double your money?

Assuming an initial investment of say £100, how many years does it take to grow to £200 given an interest rate of 5%?

Step-by-step

1. Create a new suitable method in the Lab4.
2. Create suitable variables to store the initial money, current money (at the end of each year), interest rate (5%), and years (to double the money).
3. Write code to calculate the number of years it takes to get £200.

   Tip: Use a while loop which stops when the current money == £200.

Part 4 — Nested loop practice

Ensure you can code up nested loops, understanding the full sequence in which everything runs. Effectively use the outer and 'inner' loop variables together in a nested loop. In this part you'll produce a multiplication table.

Step-by-step

1. Create a method called multiplicationTable() in the Lab.
2. We want you to produce this output on the console.

1	2	3	4	5	6	7	8	9	10
2	4	6	8	10	12	14	16	18	20
3	6	9	12	15	18	21	24	27	30
4	8	12	16	20	24	28	32	36	40
5	10	15	20	25	30	35	40	45	50
6	12	18	24	30	36	42	48	54	60
7	14	21	28	35	42	49	56	63	70
8	16	24	32	40	48	56	64	72	80
9	18	27	36	45	54	63	72	81	90
10	20	30	40	50	60	70	80	90	100

Tip: Two nested for loops (count from 1..10) are best for this.

Also, to print the product of two variables called row and col in five spaces, use a statement like:

```
System.out.printf("%5d", col * row);
```

Lab challenge — Loops and Arrays (if you have time)

Objective

The objectives of this practical are to get fully up to speed with array definition and go further than the previous lab. This lab leaves you to investigate a couple of methods without any lab instructions.

Given the following deck of cards and their values:

```
String[] cards = { "Ace Spades", "2 Spades", "3 Spades", "4 Spades", "5 Spades", "6 Spades", "7 Spades", "8 Spades", "9 Spades", "10 Spades", "Jack Spades", "Queen Spades", "King Spades", "Ace Hearts", "2 Hearts", "3 Hearts", "4 Hearts", "5 Hearts", "6 Hearts", "7 Hearts", "8 Hearts", "9 Hearts", "10 Hearts", "Jack Hearts", "Queen Hearts", "King Hearts", "Ace Clubs", "2 Clubs", "3 Clubs", "4 Clubs", "5 Clubs", "6 Clubs", "7 Clubs", "8 Clubs", "9 Clubs", "10 Clubs", "Jack Clubs", "Queen Clubs", "KingClubs", "Ace Diamonds", "2 Diamonds", "3 Diamonds", "4 Diamonds", "5 Diamonds", "6 Diamonds", "7 Diamonds", "8 Diamonds", "9 Diamonds", "10 Diamonds", "Jack Diamonds", "Queen Diamonds", "King Diamonds" };

int[] values = { 10, 2, 3, 4, 5, 6, 7, 8, 9, 10, 10, 10, 10, 10, 2, 3, 4, 5, 6, 7, 8, 9, 10, 10, 10, 10, 10, 2, 3, 4, 5, 6, 7, 8, 9, 10, 10, 10, 10, 10, 2, 3, 4, 5, 6, 7, 8, 9, 10, 10, 10, 10};
```

Draw 5 cards at random (Use the class Random).

Display the cards and their total value.

(Jack, King, Queen, and Ace have a value of 10. Other cards have the same value as the first character of their names imply).

Draw two hand and see which one is the winner!

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